

Sirui Zou

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EDUCATION

- University of California, Berkeley (UC Berkeley)**
Visiting Student: BGA Program Focus: Economics & Computational Social Science
• **Core Courses:** ECON141 (Econometrics), CogSci C131 (Computational Models of Cognition), Challenge Lab
- Peking University (National School of Development)**
Summer School: Quantitative Finance
• **Grade:** A+ (Top 10/150). Focused on stochastic analysis and quantitative pricing models.
- Southwestern University of Finance and Economics (SWUFE)**
Economics
• **Academics & Quant:** GPA: 3.9/4.0 | GRE Quant: 170/170 | GMAT FE: 655 (Top 9% Globally, Pre-enrollment)
• **Core Courses:** Macroeconomics (A+), Python (A+), Advanced Math (A), Linear Algebra (A), Probability & Statistics (A-)
• **Scholarships:** Academic Scholarship ('24, '25), Innovation & Entrepreneurship Scholarship ('24)

ACADEMIC & RESEARCH EXPERIENCE

- PersonaForge: Role-Playing Agent Research based on Dual-Process Architecture**
• **Role:** Co-first Author. **Accepted to ACL 2026 Findings**
• **Method:** Designed a three-layer personality architecture (Big Five + Defense Mechanisms + Dynamic State) with a selective dual-process generation mechanism, addressing LLM “persona drift” in long-text role-playing simulations.
• **Results:** Reduced persona drift by **75%** (6.3% vs. 24.8% baseline) across 50-turn dialogues on 88 characters; selective activation achieves **96%** of full-system performance with only **13.4%** token overhead. Validated on RoleBench with **73.2%** pairwise win-rate.
- LLM Behavioral Calibration for Economic Policy Inference**
Research Assistant (Collaborator: Kehang Zhu, Harvard; Prof. John Horton, MIT Sloan)
• **Research Question:** To what extent can LLM-simulated behavior be trusted for downstream policy inference, and what does it take to improve that trust?
• **Diagnostic Framework:** Benchmarked frontier LLMs against CHFS household microdata across four economic perception tasks, measuring distributional fit via accuracy and JSD at aggregate and subgroup levels.
• **Bias Patterns:** Identified three systematic failure modes — *Affluence Illusion* (high-income subgroup failures), *Vulnerability Prior* (overreaction to negative shocks), and *Neutral Avoidance* (persistent under-prediction of neutral responses) — motivating downstream calibration work.
- LLM-Assisted Consumption Policy Efficacy Quantification & Evaluation**
Research Assistant (Advisors: Prof. Hong Zou, Dean of School of Economics, & Dr. Han Xiao, SWUFE)
• **NLP Pipeline:** Built a two-stage text analysis pipeline on 220K+ government documents (2000–2022): (1) BERT-based binary classifier to identify consumption policy documents; (2) structured extraction of policy attributes (instrument type, target population, policy intensity).
• **Empirical Design:** Constructing household consumption indicators from CHFS microdata to enable heterogeneous analysis of policy instrument effectiveness across population subgroups.
- Basic Public Service Equalization and Common Prosperity**
Research Assistant (Advisor: Prof. Fan Wei, School of Economics, SWUFE) · Sichuan Province-Level Research Grant
• **Empirical Design:** Constructed a city-year panel (280 cities, 2012–2024) with Theil index and BPS Gini coefficients; estimated two-way fixed effects models across six BPS dimensions (education, healthcare, social security, infrastructure, culture, ecology).
• **Key Finding:** BPS inequality significantly exacerbates urban-rural income inequality across all six dimensions, supporting an “inequality → inequality” mechanism; ecological and educational BPS show the largest effects.

TECHNICAL & ENGINEERING PROJECTS

ScrollWeaver & Souloverse: Multi-Agent Social Simulation Engine *Team Lead*

Oct 2025 – Dec 2025

- **Project Overview:** Designed and developed a multi-agent social simulation engine that breathes life into static texts (novels, worldbuilding documents, lore collections), transforming them into interactive, co-creatable, and exportable living worlds. (**Global 3rd Prize, 2025 Soul AI Agent Competition**)
- **Technical Implementation:** Built a LangChain-based Orchestrator-Performer architecture with FastAPI/WebSockets for real-time streaming and D3.js for dynamic maps. Empowered multi-LLM agents (GPT, Claude) with autonomous logic and social deduction via RAG, ChromaDB/SQLite memory, a 3-layer personality model, and a dual-process cognitive architecture.

Magic Brush: AI-Powered Drawing Tutor for Children *Private Developer*

Nov 2025 – Dec 2025

- Competed in the **Doubao × Intel “Make Ideas Real” AI App Creation Challenge**; awarded **Bronze Prize (Top 30)**. Built an AI-powered step-by-step sketch drawing assistant designed to engage children independently through interactive guidance and real-time feedback.
- **Stack:** Doubao App Generation platform; multi-turn interactive dialogue flow with progressive drawing instruction and child-friendly UX design.

Course Prism: Course Evaluation & Retrieval Community *Private Developer*

Mar 2025 – Jun 2025

- **Development:** Independently developed SWUFE’s open-source course evaluation platform. Built interactive UI via Next.js/React and implemented RESTful APIs/unified auth using Django REST, PostgreSQL, and Redis.
- **Deployment:** Configured Docker Compose, Nginx reverse proxy, and Huey async task queues for one-click containerized deployment, automated data backup, and rapid recovery.

SKILLS & INTERESTS

Tools: Python (PyTorch, Pandas, NumPy), SQL, Stata, LaTeX, Linux, Docker, Git

Interests: Computational Social Science, Economics, NLP, LLM Agents

Languages: English (IELTS 7.0), Mandarin (Native)